EUROPEAN TRANSACTIONS ON ELECTRICAL POWER
Euro. Trans. Electr. Power 2010; **20**:34–51
Published online 7 October 2009 in Wiley InterScience
(www.interscience.wiley.com) DOI: 10.1002/etep.393

Evaluation of fault relative location algorithms using voltage sag data collected at 25-kV substations

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SUMMARY

Source location algorithms, recently proposed in the literature as a means to determine the origin of sags upstream or downstream from the measuring point, have been compared. Testing has been carried out using 471 records of asymmetrical voltage sags gathered from HV/MV substations of the Catalan power network, in the northeast of Spain. Six different algorithms have been included in the test, and the significance of their attributes has been analysed using a data mining approach. Multivariate statistical theory (MANOVA) has been used in this analysis. The work has been completed by improving the performance of the algorithms applying a machine learning induction algorithm (CN2), designed to extract new classification rules dealing with the combination of the attributes defined by the algorithms analysed. The study considers phase-to-ground and phase-to-phase faults separately. Copyright © 2009 John Wiley & Sons, Ltd.

KEY WORDS: power quality monitoring; voltage sag (dip) source location; fault location; position measurement

1. INTRODUCTION

Due to its impact on sensitive industrial loads and costs incurred by damage and maintenance, voltage sag analysis has captured the attention of utilities and researchers [1]. Voltage sags can be the consequence of either faults (phase-to-ground or phase-to-phase) or normal network operation (commutation of loads, discharges, relay operation, etc.) and consequently their origin can be in either the transmission or the distribution system. Depending on the origin and their propagation through the network, the alteration of voltage and current waveforms can have severe consequences on customers—e.g. the normal operation of electric appliances—and/or network maintenance—e.g. acceleration of the aging of insulators and conductors due to over currents [2,3]. Hence, the first step involved in voltage sag analysis consists of finding out if their origin is upstream or downstream of the monitoring point. This is necessary for further tasks such as fault distance estimation and stochastic analysis, commonly involved in maintenance or planning, or to assign responsibilities in cases of power quality complaints [1].

In this paper, we focus on the determination of the origin, upstream or downstream, of voltage sags collected by PQMs installed in HV/MV transformers (Figure 1) at substations. Six source location algorithms, recently proposed in the literature, have been analysed and compared. All of them analyse variations in electric parameters extracted from voltages and currents before and during the fault, and apply simple decision rules to determine the origin of sags. The significance of these parameters with respect to the origin (up/downstream) of sags has been analysed and the performance of the algorithms evaluated with real data from a data mining perspective. This comparative analysis has been carried out with records of voltage sags collected from 25-kV substations of the Catalan power system, in the northeast of Spain.

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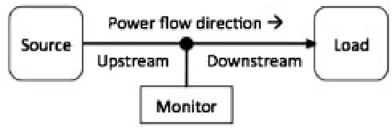

Figure 1. Voltage sag source relative location problem.

1.1. *Voltage sag source relative location algorithms*

The problem of estimating fault location from sag records has been addressed from different perspectives during the last two decades [4]. Electrical laws and statistical criteria have been the bases of the majority of methods proposed in the literature to classify voltage sags according to their origin up or downstream. Electrical laws are used to obtain simple attributes of voltage sags sensitive to origin, with typically a binary decision rule applied to discriminate between the two origins. In contrast, statistical methods usually make use of multivariate analysis to propose classifications according to the adequacy of data in statistical models obtained from previous examples in a defined confidence interval. Examples of statistical methods are presented in Refs. [2,3], while in Ref. [4] a comparison of five relative location algorithms based on electrical laws is performed with synthetic data generated in simulation. These algorithms use different concepts to discriminate between up or downstream origin: disturbance power and energy [5], slope of system trajectory (SST) [6], real current component (RCC) [7], distance relay (DR) [8] and resistance sign (RS) [9]. The comparison gave DR as the best algorithm, with RS obtaining the poorest results. More recently, a new algorithm has been proposed. This is based on the change in the phase angle of the positive-sequence component of the fault and I_{pre}s [10] (phase change in sequence current—the PCSC algorithm).

The attributes involved in the final decision rules of these algorithms are diverse and their performances when confronted with different types of faults (phase-to-ground, phase-to-phase or three-phase) can vary. In this paper, we analyse the implications of this when the algorithms are submitted to real records, in order to study their applicability in a real context. The results we obtain are slightly different from those in Ref. [4]. A possible explanation for these differences is the different natures (synthetic and real) of the data used in the two analyses. Although the algorithms being tested are the same, synthetic data are usually generated to cover theoretical problems and do not usually represent the complexity of the real world (presence of high frequency transients, noise, nonlinearity, couplings and interactions, etc.). Moreover, real data are obtained under conditions that are not perfectly known (unbounded circuits, unknown loads and impedance faults, etc.) and the adequacy of theoretical models is difficult to confirm. Therefore, the comparisons and analyses presented in this paper have to be interpreted from the perspective of applicability.

All the algorithms (Table I) analysed in this paper are suitable for radial systems (although some of them would also perform in meshed circuits) and use information extracted from three-phase voltage and current signals (except the PCSC, which only needs information from current waveforms). They are based on an analysis of steady-state change from pre-fault to fault (except the SST, which only needs information from the fault state. In Table I the required information in terms of number of cycles during the pre-fault and fault states is summarised.

1.2. *Data mining and analysis of voltage sag attributes*

The algorithms detailed in the previous subsection are based on an analysis of how electric parameters such as impedance, phase angle, magnitude, etc. or attributes extracted from the sag record, change from the pre-fault to fault states. These algorithms can be represented as simple decision rules in which Boolean conditions surrounding one of these parameters are evaluated in order to classify voltage sags according to their origin.

Table I. Relative location algorithms used in this analysis.

	Power system		Required information			
			Signals		Number of cycles	
	Radial	Meshed	Voltage	Current	Pre-fault	Fault
SST	√	√	√	√	0	All
RCC	√	√	√	√	Few	Few
DR	√	√	√	√	1	1
RS and sRS	√	×	√	√	1	Several
PCSC	√	×	×	√	1	1

DOI: 10.1002/etep

In this paper, we intend to analyse the significance of each of the attributes proposed in the algorithms using data mining multivariable analysis of variance (MANOVA), and to apply an inductive learning method to extract new rules by combining the attributes. A well-known induction algorithm called CN2 [11,12] has been used to extract rules and patterns from a set of voltage sags collected in a real environment and described by all the attributes involved in the source location algorithms analysed. The attributes described in the algorithms are computed with MatLab® and MANOVA is performed with SPSS®.

1.3. Organisation of the paper

This paper is organised in the following sections: Section 3 gives a description of the data used in this study. Section 4 gives a brief description and the performance results of each fault relative location algorithm. Section 5 analyses the features used in each algorithm by means of a descriptive statistical analysis and MANOVA. This analysis was done to assess the amount of information about the relative location contained in each feature. In Section 6, MANOVA statistical results are supported by the rules extracted using the CN2 induction algorithm. Section 7 compares the performance of the algorithms taking into account the type of fault. Finally, the main conclusions are given in Section 8.

2. VOLTAGE SAG EVENTS: DATA DESCRIPTION

The Spanish public utility, ENDESA DISTRIBUCION, provided a set of voltage sags classified as upstream and downstream (Table II), which were collected by power quality monitors installed in the secondary sides of HV/MV transformers of radial distribution networks (25-kV) in the northeast of Spain. The data set is well balanced (228 downstream and 243 upstream records), waveforms were sampled at 128 samples per cycle (50 Hz) and all the records contain 40 cycles of the three single-phase voltage and three line current signals.

Figure 2 depicts voltage sag magnitudes *versus* their duration. From this picture it can be seen that the use of duration and magnitude as discriminant attributes for classification will result in a very bad performance. Although the majority of upstream sag

Table II. Voltage sag events gathered and used in the analysis.

	Single-phase	Phase-to-phase	Total
Downstream	118	120	228
Upstream	92	151	243
Total	210	261	471

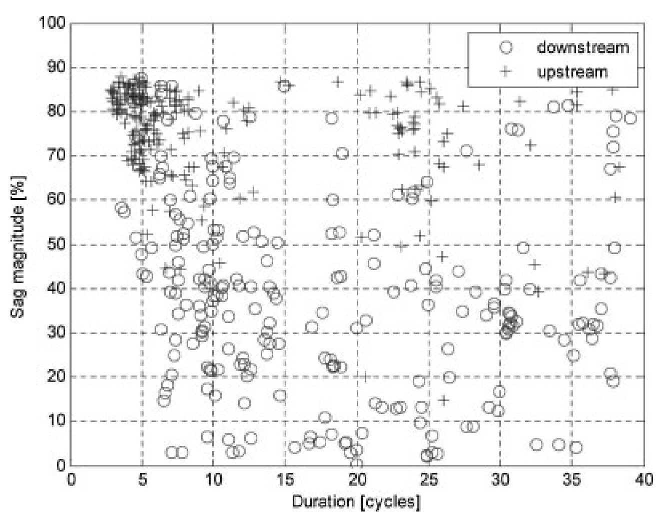

Figure 2. Magnitude of voltage sags *versus* duration.

DOI: 10.1002/etep

events last between 3 and 10 cycles, and the majority of downstream sags are of a magnitude lower than 60%, these simple rules are not reliable enough. This behaviour is quite normal because faults generated in distribution systems (downstream) are in general longer and more serious than faults generated in transmission systems (upstream).

Short Fourier Transform (SFT) with a 128-sample sliding window (one cycle) has been used to estimate the RMS voltage and current value sequences. Starting and ending times of the stationary and non-stationary parts of the sag events were obtained by applying a segmentation procedure to the time-dependant RMS value sequence, using a threshold equal to 10% of the nominal voltage [13]. For all sag events, the beginning of the first non-stationary parts was taken in the sample where the voltage drops to 90% of the pre-fault voltage.

The type of fault for each sag event was obtained using the six-phase algorithm [14], which estimates the type of fault from the three-phase voltages and the three phase-to-phase voltages. The type of fault is needed because algorithms such as the DR require the proper voltage–current pairs to compute their features and compare performance for each type.

3. DEFINITION AND RESULTS OF THE FAULT RELATIVE LOCATION ALGORITHMS

A brief description of the previously introduced algorithms is included in this section, with special attention being paid to the features used to define the decision rule that represents each algorithm. Their performance is also analysed, and the relevance of the different attributes for source location is qualitatively analysed.

3.1. Slope of the system trajectory (SST)

This algorithm is based on the fact that the relationships between the product of voltage magnitude and power factor, and the current magnitude at the measurement location, are not the same for different fault locations. For a fault located downstream of the monitoring point (Figure 1), the active power measured at the PQM is as follows [6,4]:

$$VI\cos(\theta-\alpha) = -RI^2 + E_\mathrm{s} I\cos\theta_\mathrm{s} \tag{1}$$

where E_s is the voltage at the source, R is the real part of the impedance behind the PQM, V and I are RMS voltage and current measured by the PQM, $\cos(\theta-\alpha)$ is the power factor at the PQM location and θ_s is the phase difference between the source E_s and the current I. Both parts of (1) can be divided by I to obtain (2):

$$V\cos(\theta-\alpha) = -RI + E_\mathrm{s}\cos\theta_\mathrm{s} \tag{2}$$

If $\cos(\theta-\alpha)>0$, the active power flows toward the load, the disturbance is downstream and one obtains $|V\cos(\theta-\alpha)| = V\cos(\theta-\alpha)$. This relationship corresponds to a line equation with slope $-R$, as shown in (3):

$$|V\cos(\theta-\alpha)| = -RI + E_\mathrm{s}\cos\theta_\mathrm{s} \tag{3}$$

If the fault is upstream, a linear equation with slope $+R$ is obtained, as shown in (4):

$$|V\cos(\theta-\alpha)| = +RI - E_\mathrm{s}\cos\theta_\mathrm{s} \tag{4}$$

The SST algorithm assumes that $\cos\theta_\mathrm{s}$ does not greatly change during the disturbance, and that therefore the slope of the line fitting the points of (I, $|V\cos(\theta-\alpha)|$) during the voltage sag will be negative for a downstream event, and positive for an upstream one. Hence, the decision rule for the SST algorithm can be denoted as follows [6]:

'IF Slope[I, $|V\cos(\theta-\alpha)|$] <0 THEN downstream ELSE upstream'.

In this work, the slope was computed using the single-phase voltage with the lowest magnitude. The samples between the beginning of the sag (first segment) and the beginning of the second non-stationary part (third segment) were taken to do the computation.

Figure 3 depicts the sag relative locations estimated using the SST algorithm. The horizontal axis represents each sag event, while the vertical axis represents the slope of the values I and $|V\cos(\theta-\alpha)|$. The sag events whose relative location is downstream are depicted in the first 228 positions of the horizontal axis and upstream sags are depicted between positions 229 and 471.

According to the SST algorithm rule, the downstream sag events would fall inside the bottom left shaded region (the negative slope), while the upstream sag events would be inside the upper right shaded region (the positive slope).

DOI: 10.1002/etep

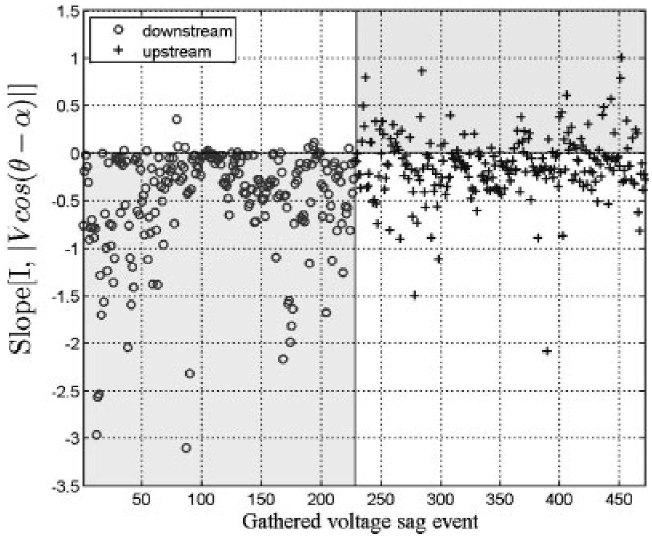

Figure 3. Slope of system trajectory algorithm results. The downstream sag events (circles) are depicted in the first 228 positions of the horizontal axis.

Notice that the SST algorithm classifies the downstream sags better than the upstream ones, since many of them are depicted inside the downstream region. Conversely, less than half the upstream sags are inside the upstream region.

3.2. Real current component (RCC)

The RCC algorithm [7] uses the polarity of the RCC to determine the relative location of the sag source. The product of the RMS current and power factor at the PQM point is employed to locate the sag source. This approach starts by extracting the real part of Equation (3). The RCC is based on the fact that $I\cos(\theta-\alpha)>0$ for a fault whose source is located downstream.

At the beginning of the voltage sag, the current is significantly higher than the steady-state current due to the sudden change in the electrical conditions. Therefore, a more suitable feature for choosing the relative location of the voltage sag source will be based on the direction of the current at the beginning of the fault [7,6]. The decision rule is the following:

'IF I $\cos(\theta-\alpha)>0$ THEN downstream ELSE upstream'.

The polarity was computed as the integral of the product $I\cos(\theta-\alpha)$ from the beginning of the sag until the end of its first non-stationary part. The single-phase voltage with the lowest magnitude is also used in this algorithm.

The RCC rule indicates that the downstream sag events have to be depicted in the upper left shaded region ($I\cos(\theta-\alpha)>0$), whereas the upstream sag events will be in the bottom right shaded region ($I\cos(\theta-\alpha)<0$) (Figure 4).

Results are similar to those obtained with the SST (downstream sags are classified better than upstream sags). But the feature used in the RCC algorithm, $I\cos(\theta-\alpha)$, has more variability than the SST feature, Slope(I,$|V\cos(\theta-\alpha)|$).

3.3. Distance relay (DR)

This algorithm is based on the principle that the magnitude and angle of impedance before and after the sag event clearly indicate the relative location of the sag source with respect to the PQM point [8]. For this purpose, the DR algorithm uses DR information for sag source detection. For the downstream fault in the system shown in Figure 1, the impedance seen at the PQM point will be:

$$Z_{\mathrm{PQM}}=\frac{V\angle\theta}{I\angle\alpha}=Z'+\Delta Z \tag{5}$$

where Z' is impedance up to the fault point and ΔZ is a function of fault resistance, load angle, etc. In the case of a fault behind the PQM point, the current direction will be reversed and the resulting impedance will change in both magnitude and angle. Hence, for a

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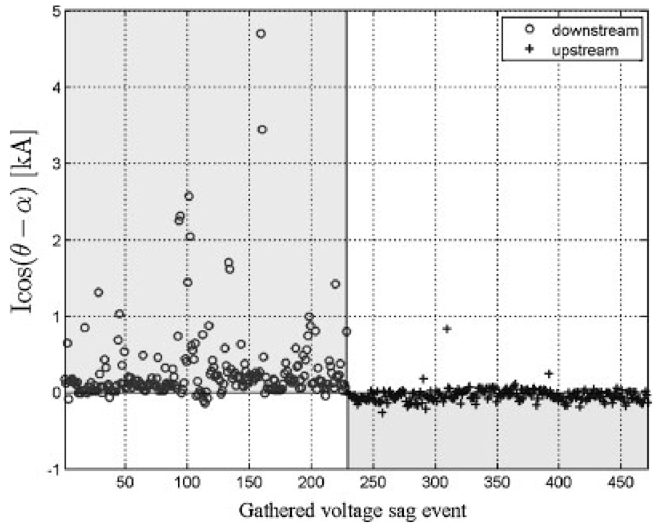

Figure 4. Real current component algorithm results. The downstream sag events (circles) are depicted in the first 228 positions of the horizontal axis.

downstream fault the impedance seen during the fault (Z_{sag}) will decrease in respect of the impedance seen in steady state (Z_{ss}) and its phase angle will increase. The DR rule for sag source detection is:

'IF $Z_{ratio} < 1$ AND $\angle Z_{sag} > 0$ THEN downstream ELSE upstream'.

According to the different types of faults, the Z_{ratio} is the ratio between $|Z_{sag}|$ and $|Z_{ss}|$, and the proper voltage–current pair must be used in estimating them [8]. In this work, the proper pairs were obtained using the six-phase algorithm [14].

According to the DR algorithm rule, the downstream sag events will be plotted in the shaded bottom right region ($Z_{ratio} < 1$ and $\angle Z_{sag} > 0$), as shown in Figure 5, while the upstream sag events will be outside the shaded region.

Figure 5 clearly shows that the DR algorithm adequately discriminates between the two types of sag relative location. Notice that upstream sags are classified better than downstream sags. Only 1 upstream sag was misclassified while 24 downstream sags were considered as upstream.

Another important observation relates to the independent performance of Z_{ratio} and $\angle Z_{sag}$ features. It can be seen in Figure 5 that Z_{ratio} is able to discriminate between upstream and downstream sags without $\angle Z_{sag}$ because most of upstream sags have Z_{ratio} values

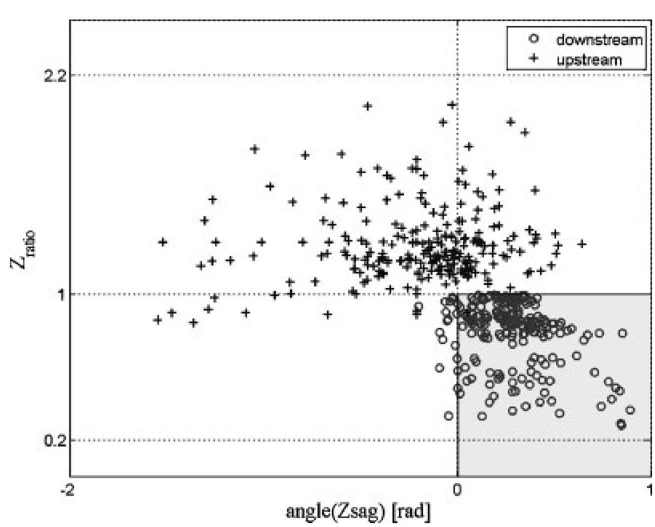

Figure 5. Distance relay algorithm results. Sags are classified as downstream sags if $Z_{ratio} < 1$ and $\angle Z_{sag} > 0$.

DOI: 10.1002/etep

greater than 1, while downstream sags have values lower than 1. Conversely, the greater part of downstream sags and a reasonable number of upstream sags have $\angle Z_{\mathrm{sag}}$ greater than zero, meaning that $\angle Z_{\mathrm{sag}}$ is not better at discriminating voltage sags than Z_{ratio}. We can therefore conclude that the Z_{ratio} feature contains more information about the sag relative location than $\angle Z_{\mathrm{sag}}$. The DR algorithm gives better results if both features in the same condition rule are used.

3.4. Resistance sign (RS)

This algorithm is based on the principle of estimating the equivalent impedance of the non-disturbance side by utilising the voltage and current changes caused by the disturbance. A sign of the real part of the estimated impedance can reveal if the sag event is from upstream or downstream.

For a downstream fault from the PQM point in Figure 1 the voltage is as follows:

$$V = E_{\mathrm{S}} - IZ \tag{6}$$

where V and I are the voltage and current at the PQM point and Z is the impedance behind the PQM point. In order to improve the estimate of the impedance, the authors propose utilising multiple voltage and current cycles by means of the least-squares (LS) method. Equation (6) can be written as a function of real and imaginary parts as follows:

$$V_{\mathrm{X}} + \mathrm{j}V_{\mathrm{Y}} = (E_{\mathrm{SX}} + E_{\mathrm{SY}}) - (I_{\mathrm{X}} + \mathrm{j}I_{\mathrm{Y}})(R + \mathrm{j}X) \tag{7}$$

where X and Y represent real and imaginary parts of each variable, respectively. If n pre-fault and fault cycles of (V, I) data are measured during the sag event, the equivalent impedance $(R + \mathrm{j}X)$ can then be found using the LS method. This means that equivalent impedance can be computed through two expressions, the first (8) based on the real part of the voltage values, and the second (9) on the imaginary part

$$\begin{bmatrix} R \\ X \\ E_{\mathrm{SX}} \end{bmatrix} = \begin{bmatrix} I_{\mathrm{X}}(1) & I_{\mathrm{Y}}(1) & 1 \\ \vdots & \vdots & \vdots \\ I_{\mathrm{X}}(n) & I_{\mathrm{Y}}(n) & 1 \end{bmatrix}^{\oplus} \begin{bmatrix} V_{\mathrm{X}}(1) \\ \vdots \\ V_{\mathrm{X}}(n) \end{bmatrix} \tag{8}$$

and

$$\begin{bmatrix} R \\ X \\ E_{\mathrm{SY}} \end{bmatrix} = \begin{bmatrix} I_{\mathrm{Y}}(1) & I_{\mathrm{X}}(1) & 1 \\ \vdots & \vdots & \vdots \\ I_{\mathrm{Y}}(n) & I_{\mathrm{X}}(n) & 1 \end{bmatrix}^{\oplus} \begin{bmatrix} V_{\mathrm{Y}}(1) \\ \vdots \\ V_{\mathrm{Y}}(n) \end{bmatrix} \tag{9}$$

where the symbol '$\oplus$' indicates the pseudo-inverse of the matrix. The number of cycles n is determined by the power flow, so voltages and currents used in (8) and (9) must only include values before the reversion of the power flow. The authors claim that if the fault is located downstream, the equivalent resistance (R) will be negative in both Equations (8) and (9). Conversely, if both signs are positive, the sag source is upstream. If the signs are different (opposite), the test is not conclusive. The RS rule is as follows:

'IF $R_{\mathrm{ex}} > 0$ AND $R_{\mathrm{ey}} > 0$ THEN upstream ELSE IF $R_{\mathrm{ex}} < 0$ and $R_{\mathrm{ey}} < 0$ THEN downstream ELSE not conclusive test' where R_{ex} and R_{ey} represent equivalent resistance R based on real (Equation 8) and imaginary (Equation 9) parts of the voltage, respectively.

The RS algorithm rule indicates that downstream sag events will be depicted in the bottom left shaded region ($R_{\mathrm{ex}} < 0$ and $R_{\mathrm{ey}} < 0$), whereas the upstream sag events will be depicted in the upper right shaded region ($R_{\mathrm{ex}} > 0$ and $R_{\mathrm{ey}} > 0$), (Figure 6).

Most upstream sag events (242) were correctly classified. Only one was confused and classified as a downstream event. Hence, R_{ex} and R_{ey} are able to discriminate between sag events whose relative location is upstream. However, while a reasonable percentage of downstream sags (155) was correctly identified, others (32) were classified as 'Not Conclusive Test' because the sign of the resistances was different. In Ref. [4] similar results (excessive 'Not Conclusive Test' outputs) were obtained for the RS algorithm using synthetic data.

A simplified version of the RS algorithm (sRS) examines the sign of only one resistance, named Re. The simplification is obtained by applying a rotating transformation to the R_{ex} and R_{ey} expressions. The corresponding sRS decision rule is:

'IF Re > 0 THEN upstream ELSE downstream'.

Downstream sag events will be depicted in the shaded bottom left region (Re < 0), and the upstream sag events will be depicted in the shaded upper right region (Re > 0), (Figure 7).

DOI: 10.1002/etep

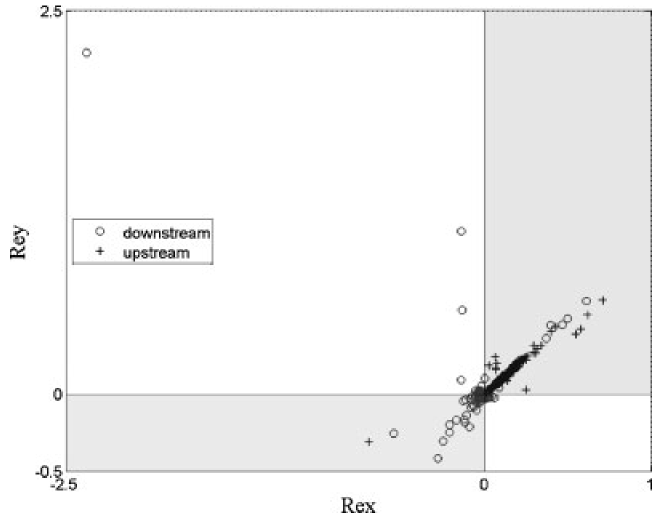

Figure 6. Resistance sign algorithm results. Voltages sags are classified as downstream sags if $R_{ex} < 0$ and $R_{ey} < 0$.

The performance of the Re feature is worse than that of R_{ex} and R_{ey}. It can be seen that a high percentage of upstream sag events (218) are correctly classified whereas several downstream sags (74) are misclassified.

3.5. Phase change in sequence current (PCSC)

This estimates the sag relative location using the difference in phase angle between the fault and pre-fault positive-sequence component of the current. The positive sequence is used because it is available for all types of faults. Figure 8 shows the phasor diagram of the power network shown in Figure 1, where I_{up} and I_{down} are currents at the PQM point for upstream and downstream faults, I_{pre} corresponds to the current before the sag event, and $\Delta\phi_{up}$ and $\Delta\phi_{down}$ are the difference in phase angle between the fault currents (I_{up}, I_{down}) and the pre-fault current (I_{pre}). V_s and V_L are the voltages at source and load, respectively.

From Figure 8 it can be inferred that the sag source relative location can be identified from the fault current phasor position in relation to the I_{pre} phasor position. As shown in the phasor diagram, the angle difference of the fault and I_{pre} phasor for upstream faults is positive ($\Delta\phi_{up}$), and for downstream it is negative ($\Delta\phi_{down}$). The PCSC rule is [10]:

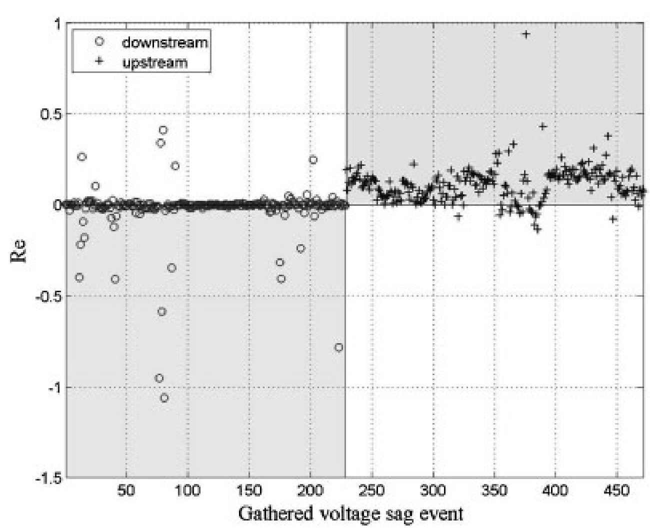

Figure 7. Simplified resistance sign algorithm results. Voltage sags are classified as downstream sags if Re < 0.

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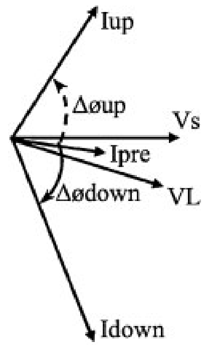

Figure 8. Phasor diagram of the network shown in Figure 1.

'IF $\Delta\phi < 0$ THEN downstream ELSE upstream'.

One cycle before the fault is used to compute the I_{pre} phasor and another cycle after the fault inception is used for the estimation of the fault phasor.

According to the PCSC rule, downstream sag events will be plotted in the bottom left shaded region ($\Delta\phi < 0$), while the upstream sag events (crosses) will be inside the upper right shaded region ($\Delta\phi > 0$).Figure 9 depicts the variation in the phase change in sequence currents ($\Delta\phi$) with respect to the set of sag events. It can be clearly seen that $\Delta\phi$ has values between $-\pi$ and zero for downstream sags and between zero and π for upstream sag events. Only a few sag disturbances are misclassified (17). This shows that the $\Delta\phi$ feature is clearly affected by the sag source relative location.

We can summarise these findings by noting that the features with the best performance were Z_{ratio} (DR), R_{ex}–R_{ey} (RS) and $\Delta\phi$ (PCSC). The performance of Slope$(I,|V\cos(\theta-\alpha)|)$ (SST), $I\cos(\theta-\alpha)$ (RCC) and Re (RS) was moderate, while that of $\angle Z_{sag}$ (DR) was the poorest. This is shown in Table III.

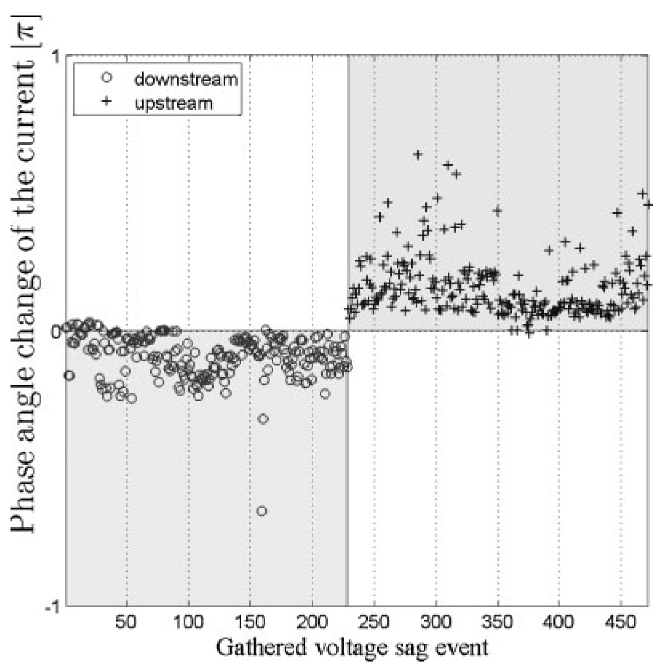

Figure 9. Phase change in sequence current algorithm results. Downstream sag events (circles) must be inside the shaded bottom left region ($\Delta\phi < 0$).

DOI: 10.1002/etep

Table III. Qualitative performance of each feature.

Algorithm	Qualitative performance		
	Good	Moderate	Poor
SST			$\mathrm{Slope}(I,\lvert V\cos(\theta-\alpha)\rvert)$
RCC			$I\cos(\theta-\alpha)$
DR	Z_{ratio}		$\angle Z_{\mathrm{sag}}$
RS	R_{ex}, R_{ey}		
sRS		Re	
PCSC	$\Delta\phi$		

4. FEATURE ANALYSIS

In the previous section, we have seen that some attributes are more sensitive to the origin of sags than others. The purpose of this section is to quantify the significance of this and extract patterns and characteristics that might help us to better interpret and analyse the available data set with respect to these attributes. The specific technique used is the MANOVA, which allows us to determine which features are relevant to sag relative location. Thereafter, the machine learning inductive algorithm CN2 is applied to voltage sags described by the selected features to automatically extract rules that describe the data (voltage sags) according to their origin. A combination of logic conditions over the extracted attributes appears in the antecedents, while the conclusion of the extracted rules is the origin of sags. The initial set of attributes is summarised in Table V, with a brief definition of each one.

4.1. Outlier correction

Since inductive algorithms are sensitive to outliers, an outlier analysis was performed to avoid systematic mistakes due to outlier values. As a result, the furthest voltage sag events from the mean of each feature were modified. Therefore, outlier sags were not excluded. The outlier correction was performed with confidence interval of 95%. It consists of replacing 5% of the highest (2.5%) and lowest (2.5%) values for each feature by the mean value of the corresponding feature. Based on the amount of upstream and downstream voltage sags (Table II), the six highest and six lowest values for each feature were replaced. The outlier correction was performed on each class (upstream and downstream). Finally, 192 elements from the voltage sag data matrix (471×8) were modified, corresponding to 5.0% of the elements of the data matrix. Finally, the selected features after outlier correction were used as input for the CN2 algorithm.

4.2. Descriptive statistical analysis

In Table IV the mean, μ, and standard deviation, ∂, of each feature in each class are listed (data with the corrected outliers).

Analysis of the mean and standard deviation values of each class in the $\mathrm{Slope}(I,\lvert V\cos(\theta-\alpha)\rvert)$, $I\cos(\theta-\alpha)$ and $\angle Z_{\mathrm{sag}}$ features show that the downstream and upstream class overlap. For instance, taking the feature $I\cos(\theta-\alpha)$, the mean minus one standard deviation of the downstream class is equal to -38.7, while the mean plus one standard deviation of the upstream class is equal to 21.69. Hence, although the centres of the classes are not close, with only one standard deviation they overlap. As a result, the aforementioned features obtain a low performance (Table III). There are no overlapping classes in the rest of the features.

4.3. Multivariate analysis of variance—MANOVA

The main purpose of MANOVA is to explore how independent variables influence the patterning of response in dependent variables. Thus, MANOVA allows the following question to be answered: what is the importance of each feature in the source relative location? From this, we can determine the influence grade of the source location in each feature. The sag source relative location (upstream or downstream sag) was used as the independent variable and the $\mathrm{Slope}(I,\lvert V\cos(\theta-\alpha)\rvert)$, $I\cos(\theta-\alpha)$, Z_{ratio}, $\angle Z_{\mathrm{sag}}$, R_{ex}, R_{ey}, Re and $\Delta\phi$ features were used as dependent variables.

Table V shows the quality of the sag source location effect for each feature. Quality values near 100% indicate that most of the variability in this feature is associated with the source relative location, while values near 0% indicate that the feature does not contain information about sag source relative location. The quality values listed in Table V show that the features with the most

DOI: 10.1002/etep

Table IV. Feature descriptive statistics.

Feature	Algorithm	Class	μ	∂	Overlapping class
SlopeIV	SST	Downstream	−0.442	0.433	Yes
		Upstream	−0.128	0.241	
$I\cos(\theta-\alpha)$	RCC	Downstream	238.1	276.8	Yes
		Upstream	−34.81	56.50	
Z_{ratio}	DR	Downstream	0.794	0.174	No
		Upstream	1.270	0.209	
$\angle Z_{\text{sag}}$	DR	Downstream	0.278	0.194	Yes
		Upstream	−0.211	0.406	
R_{ex}	RS	Downstream	−0.014	0.032	No
		Upstream	0.137	0.057	
R_{ey}	RS	Downstream	−0.004	0.049	No
		Upstream	0.139	0.056	
Re	sRS	Downstream	−0.015	0.050	No
		Upstream	0.106	0.068	
$\Delta\phi$	PCSC	Downstream	−0.277	0.192	No
		Upstream	0.467	0.269	

information are Z_{ratio}, R_{ex}, R_{ey} and $\Delta\phi$, respectively. Conversely, the medium and lower quality features were Re, Slope$(I,|V\cos(\theta-\alpha)|)$, $I\cos(\theta-\alpha)$ and $\angle Z_{\text{sag}}$.

The Slope$(I,|V\cos(\theta-\alpha)|)$–$I\cos(\theta-\alpha)$ and R_{ex}–R_{ey} pairs are equally significant with respect to sag relative location—about 44 and 80% for each pair, respectively.

Similarly, Table V confirms that the Z_{ratio} feature contains more information than $\angle Z_{\text{sag}}$, as pointed out in the previous section. In relation to the RS algorithm, R_{ex} and R_{ey} are a little more significant than the Re feature used in its simplified approach.

We can conclude that, in general, MANOVA results are similar to the qualitative classification listed in Table III for all the features.

5. COMBINATION OF FEATURES TO IMPROVE SAG SOURCE LOCATION

In this section, we define the CN2 algorithm and explain the tasks performed to extract the sag source location rule set. These rules are tested with the same sag events used to test the algorithms. After that, in the next section, rule classification results are compared with the algorithm results.

5.1. The CN2 induction algorithm

The CN2 is a rule extraction algorithm that induces an ordered list of classification rules from a set of labelled observations [11,12]. The rules are in the form Condi $\rightarrow$ Classj, where the property of interest is Classj that appears in the rule consequent, and the condition Condi is a conjunction of the features selected from training observations. The CN2 works as an iterative process; each iteration forms a condition that covers a large number of observations of a single Classj and a few observations of other classes.

Table V. Quality of the source relative location effect over the feature.

Feature	Definition	Algorithm	Quality (%)
Slope$(I,|V_{\text{pf}}|)$	Slope of the I and $|V\cos(\theta-\alpha)|$ values	SST	45.9
$I\cos(\theta-\alpha)$	Product of the RMS real current and power factor angle at the beginning of the sag	RCC	41.9
Z_{ratio}	Ratio of fault impedance to steady-state impedance	DR	96.8
$\angle Z_{\text{sag}}$	Phase angle of the impedance during the voltage sag	DR	36.7
R_{ex}	Real part of the estimated impedance from the real part of the sequence components	RS	81.8
R_{ey}	Real part of the estimated impedance from the imaginary part of the sequence components	RS	78.0
Re	Equivalent positive-sequence impedance	sRS	62.2
$\Delta\phi$	Difference in phase angle between currents during fault and pre-fault conditions	PCSC	73.1

Having found a good Condi, the algorithm removes those observations it covers from the training set and adds the rule '*If <Condi> Then Classj*' to the end of the rule list. This process iterates until no more satisfactory conditions can be found. The main CN2 parameters are as follows:

(1) Examples to cover: the percentage of examples to cover.
(2) Beam width: the number of best rules that are, in each step, further specialised. Other rules are discarded.
(3) Evaluation function: the CN2 has three evaluation functions—Entropy, Laplace and WRACC. During the search procedure, the CN2 must evaluate the rules it finds to decide which is best. In order to do so, it uses the evaluation function [11,12].

In this paper, the CN2 was implemented using Orange data mining software available online.

5.2. Experimentation and results

The point of applying the CN2 algorithm to voltage sag features is to extract the set of rules that best describe the voltage data set we have. Afterwards, the meaning of the rules is analysed.

Since the CN2 algorithm only works with discretised data, the original data set was discretised while taking into account for each feature the evaluation conditions that appear in the corresponding rules, i.e. preserving the electric meaning of those attributes with respect to the classification problem under consideration. All the features were discretised using zero as a cut point, except the Z_{ratio}, whose cut point was the unit.

The CN2 algorithm parameters used were: percentage examples to cover equal to 0.95; beam width equal to 0.5; and the WRACC evaluation function. Finally, the rules listed in Table VI were generated.

Table VI shows the rules extracted by the CN2 induction algorithm. The first and second columns correspond to the antecedent and conclusion of the rule, respectively, and the third and fourth columns are the voltage sag proportion that satisfies the condition described by the rule. The first rule describes the upstream class and covers 100 and 1.3% of the upstream and downstream sags, respectively.

It can be seen that the first rule corresponds to a combination of the PCSC ($\Delta\phi$) and RS (R_{ex}, R_{ey}) algorithms, while the second rule corresponds directly to the PCSC algorithm. Both performed well according to previously obtained qualitative (Table III) and statistical (Table V) results.

5.3. Interpretation of the extracted rules

From the extracted rules, it is possible to define a new sag relative location algorithm, which we have called the PCSC&RS rule set. The rule is as follows:

'IF $\Delta\phi \geq 0$ & $R_{\mathrm{ex}} \geq 0$ & $R_{\mathrm{ey}} \geq 0$ THEN upstream ELSE IF $\Delta\phi < 0$ THEN downstream ELSE not conclusive test'.

It can be seen that the PCSC&RS algorithm (Figure 10) is able to discriminate the upstream sag from the first quadrant of the RS space (Figure 6) and positive axes of the PCSC space (Figure 7). In other words, the PCSC&RS algorithm describes the upstream sags as the union between PCSC and RS upstream conditions, whereas downstream sags are better represented using only the PCSC downstream condition.

6. COMPARISON OF ALGORITHMS

In this section a comparison of the classification results of the six algorithms, together with the results obtained from the combination of the PCSC and RS algorithms, are presented. The results are compared according to the type of fault, using the original data without corrected outliers. A confusion matrix is used to compare the results.

Table VI. Extracted rule set using the CN2 induction algorithm.

Condition	Class	Downstream coverage (%)	Upstream coverage (%)
$\Delta\phi \geq 0$ and $R_{\mathrm{ex}} \geq 0$ and $R_{\mathrm{ey}} \geq 0$	Upstream	1.3	100
$\Delta\phi < 0$	Downstream	95.6	0

DOI: 10.1002/etep

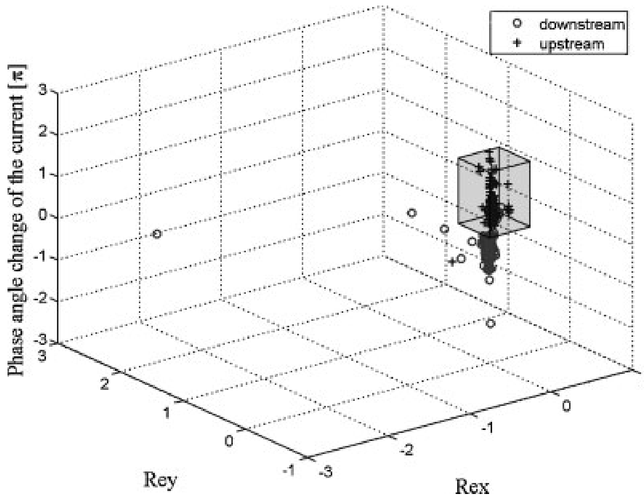

Figure 10. Combination of the PCSC and RS algorithms. Upstream sag events (crosses) have to be inside the cube.

6.1. Confusion matrix

In order to compare the rule and algorithms results obtained, a confusion matrix was used. A confusion matrix is a representation of the classification results as presented in Table VII. It shows the differences between the true and predicted classes for a set of labelled examples [15].

The evaluation of these indices allows several performance parameters of the classifier to be computed. Special attention is paid to the true positive rate (TPR) and the false negative rate (FNR):

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{10}$$

$$\mathrm{FNR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}} \tag{11}$$

In a two-dimensional graph, where the y axis represents the TPR and x axis represents the FPR, the lower left point (FPR $=0$, TPR $=0$) represents a classifier that never correctly classifies cases represented by the model. The upper left point (FPR $=0$, TPR $=1$) represents the perfect classifier (it never misclassifies cases, and, what is more, determines correctly cases that are not represented by the model). The upper right point (FPR $=1$, TPR $=1$) represents classifiers that always correctly classify cases fitting the model, but always classify incorrectly cases different from the model [16].

In general, the classifier located nearest to point (FPR $=0$, TPR $=1$) will be the best in a set of classifiers.

Table VII. Confusion matrix.

		Real class	
		Downstream	Upstream
Predicted	Downstream	TP	FP
Class	Upstream	FN	TN

TP, true positive: cases correctly predicted as the reference class (downstream); TN, true negative: cases correctly classified as the non-reference class; FP, stands for false positive: cases classified as the reference class, but their real class is the non-reference class; FN, stands for false negative: cases classified as a non-reference class, but their real class is the reference class.

DOI: 10.1002/etep

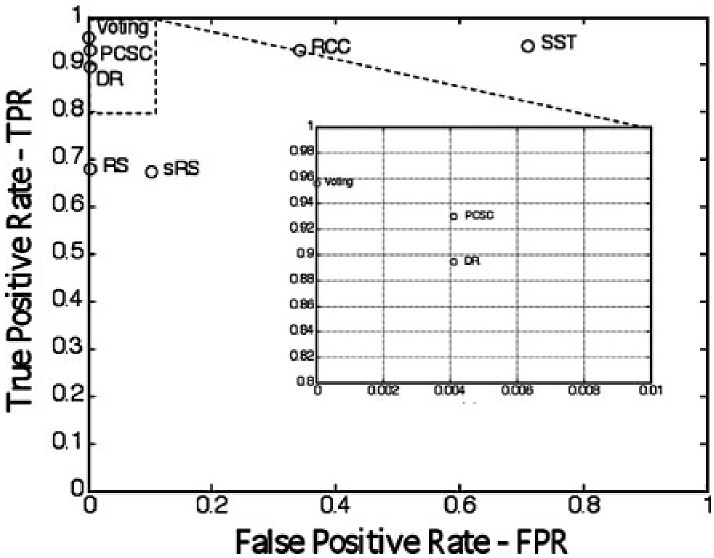

Figure 11. FPR *versus* TPR. Single-phase and phase-to-phase sag events.

6.2. Comparison

Figure 11 shows the FPR and TPR indices for each algorithm and both asymmetrical fault types (single-phase and phase-to-phase), while Figures 12 and 13 show the same indices separately computed for single-phase and phase-to-phase fault types, respectively. In Table VIII, the confusion matrices and the values taken for the indices in each algorithm and scenario are listed. The number of sags classified as not conclusive test (NCT) and the error and classification rates in percentage terms (Hit%) are also listed.

To these figures we have added another classifier called 'Voting'. The Voting classifier gives the sag source location using a democratic combination of results coming from the six algorithms described in Section 4. Its estimation will be upstream if four of the six algorithms give upstream as a result, with a similar condition applying for a downstream estimation. In other cases (where the majority is lower than four), the estimation will be marked as NCT.

The FPR and TPR indices obtained with the PCSC&RS rule set are equal to the PCSC indices. The differences between the two are seen in the FN and TN indices (Table VIII).

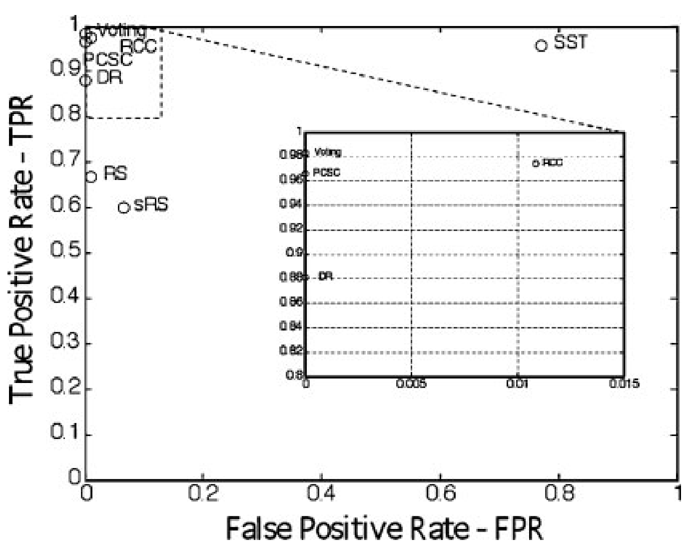

Figure 12. FPR *versus* TPR. Single-phase sag events only.

DOI: 10.1002/etep

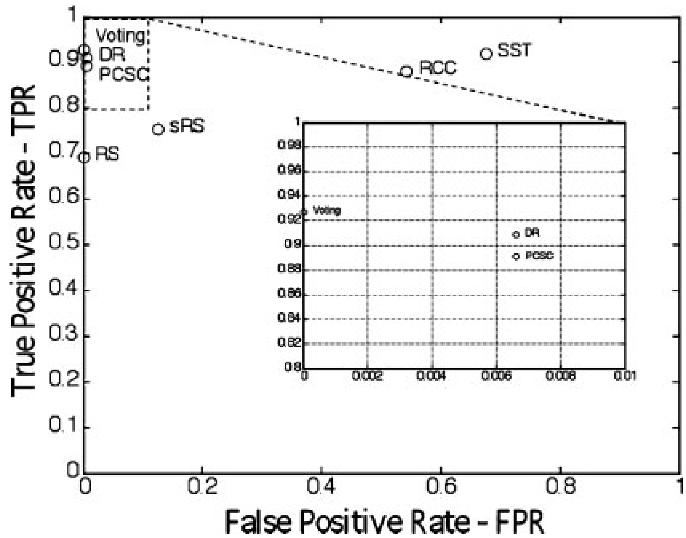

Figure 13. FPR *versus* TPR. Phase-to-phase sag events only.

6.2.1. Scenario with all sag events. If we analyse Figure 11 and Table VIII we can see that out of the global performances of each algorithm, the PCSC (0.004, 0.930) and DR (0.004, 0.895) performed best after the Voting classifier (0, 0.956) because they are nearer to the upper right point than the other algorithms. Their classification rates were 96.4 and 94.7%, respectively.

The RS (0.004, 0.680) and sRS (0.103, 0.675) algorithms have similar results with respect to downstream sag events, but in relation to upstream sags, the RS obtained the best result because the sRS misclassified 24 more upstream sags than the RS algorithm.

Table VIII. Confusion matrix and classification rates for each algorithm.

	TP	FN	FP	TN	NCT	FPR	TPR	Error%	Hit%
All (471) sags									
PCSC	212	16	1	242	0	0.004	0.93	3.6	96.4
RS	155	41	1	242	32	0.004	0.68	8.9	84.3
DR	204	24	1	242	0	0.004	0.895	5.3	94.7
RCC	212	16	83	160	0	0.342	0.93	21.0	79.0
SST	214	14	173	70	0	0.712	0.939	39.7	60.3
sRS	154	74	25	218	0	0.103	0.675	21.0	79.0
Voting	218	2	0	228	23	0	0.956	0.4	94.7
PCSC&RS	212	6	1	241	11	0.004	0.93	1.5	96.2
Single-phase (210) sags									
PCSC	114	4	0	92	0	0	0.966	1.9	98.1
RS	79	18	1	91	21	0.011	0.669	9.1	81.0
DR	104	14	0	92	0	0	0.881	6.7	93.3
RCC	115	3	1	91	0	0.011	0.975	1.9	98.1
SST	113	5	71	21	0	0.772	0.958	36.2	63.8
sRS	71	47	6	86	0	0.065	0.602	25.2	74.8
Voting	116	0	0	92	2	0	0.983	0.0	99.0
PCSC&RS	114	0	0	91	5	0	0.966	0.0	97.6
Phase-to-phase (261) sags									
PCSC	98	12	1	150	0	0.007	0.891	5.0	95.0
RS	76	23	0	151	11	0	0.691	8.8	87.0
DR	100	10	1	150	0	0.007	0.909	4.2	95.8
RCC	97	13	82	69	0	0.543	0.882	36.4	63.6
SST	101	9	102	49	0	0.676	0.918	42.5	57.5
sRS	83	27	19	132	0	0.126	0.755	17.6	82.4
Voting	102	2	0	136	21	0	0.927	0.8	91.2
PCSC&RS	98	6	1	150	6	0.007	0.891	2.7	95.0

DOI: 10.1002/etep

Although the RS algorithm uses R_{ex} and R_{ey} as features, which were assessed as good in the multivariable statistical analysis with qualities of 81.8 and 78%, respectively, RS classification rates give only average results due to the number of NCTs.

The RCC (0.342, 0.93) and SST (0.712, 0.939) algorithms adequately discriminate downstream sags but incorrectly classify a large number of upstream sag events. For instance, the SST misclassified 173 out of 243 upstream sags (72% approximately).

6.2.2. Scenario with single-phase sag events. In this scenario (Figure 12), we can see that the RCC (0.011, 0.975) algorithm is able to correctly discriminate the sags whose cause is a single-phase fault, even a little better than the PCSC algorithm (0, 0.966), because the RCC point is the nearest to the (0, 1) point after the Voting classifier. The other algorithms plot around the same zone in Figure 11.

With single-phase faults, the SST algorithm still has a low classification rate. In this scenario, 71 out of 92 single-phase upstream sags were misclassified (77% approximately).

6.2.3. Scenario with phase-to-phase sag events. In Figure 13, we can observe that the DR algorithm (0.007, 0.909) is a little better at determining phase-to-phase sags than the PCSC (0.007, 0.891), and that the sRS algorithm (0.126, 0.755) is slightly better than the RS (0, 0.691). It should be noted that the RCC algorithm classification rate is low with sag events caused by phase-to-phase faults. The SST algorithm does not improve its classification rate in this scenario.

6.2.4. Discussion of scenarios. The global results depicted in Figure 11 show that the PCSC algorithm is the best method by which to determine the voltage sag source relative location in asymmetrical conditions (96.4%), and the DR algorithm is the second best (94.7%). Moreover, application of the PCSC does not require the type of fault to be determined. For these reasons, in an online sag-monitoring project we would recommend using the PCSC algorithm.

The RS and sRS algorithms perform similarly, at 84.3 and 79%, respectively. Both algorithms describe upstream sag events better than downstream ones.

The RCC algorithm is the best at determining voltage sags caused by single-phase faults, but it is not the best in relation to phase-to-phase sags. In this case, the DR and PCSC algorithms are the best.

The performance of SST algorithm was poor with respect to upstream voltage sags. The classification indices associated with upstream sags were low in all scenarios. However, the SST algorithm achieved a high classification rate in relation to downstream sag events, correctly determining 93.9, 95.8 and 91.2%, respectively, in the three scenarios.

Although the PCSC&RS classification indices are the same as those of the PCSC algorithm, the PCSC&RS has a lower percentage of error (1.5%) than the PCSC algorithm (3.6%), because the PCSC&RS is able to distinguish sags that cannot be classified. The combination of both methods increases the reliability of the PCSC&RS rule. The PCSC&RS algorithm is more reliable than the RS and PCSC in terms of success ratios. For instance, the PCSC misclassified 11 downstream sags but the PCSC&RS was able to classify the same 11 sags as NCT.

The Voting classifier always obtained the best results. This classifier was used to illustrate how classification performance can be increased, but the approach is not practical for an online implementation.

6.3. Misclassified voltage sags

Based on the previous results, the sag events incorrectly classified by the PCSC algorithm and PCSC&RS rule set have been analysed.

The features relating to the PCSC and PCSC&RS are listed in Table IX for the 17 voltage sags misclassified by the PCSC algorithm. The real class of each one is shown in the column labelled class. A cross (x) indicates that the sag was incorrectly classified, and NCT indicates that the event was classified as a NCT.

The PCSC algorithm classified 16 downstream sags as upstream and one upstream sag as a downstream event. For downstream sags, $\Delta\phi$ should be lower than zero ($\Delta\phi < 0$), and greater than zero ($\Delta\phi > 0$) for upstream events, but it can be clearly seen that for all 17 sags the $\Delta\phi$ feature is very close to zero. All of them have been misclassified by a few hundredths and thousandths of radian units of the $\Delta\phi$ feature.

Likewise, the PCSC&RS rule set behaves the same with this group of sags, since with most of them the R_{ex} and R_{ey} features are very close to zero too. Consequently, the PCSC&RS rule set misclassified some sags and others were classified as NCT.

DOI: 10.1002/etep

Table IX. Voltage sags misclassified using the PCSC algorithm.

ID	$\Delta\phi$	R_{ex}	R_{ey}	Class	PCSC	PCSC&RS
1	0.030	0.002	0.002	Down	x	x
2	0.038	0.003	0.003	Down	x	x
11	0.070	−0.002	−0.002	Down	x	NCT
17	0.077	−0.014	−0.022	Down	x	NCT
18	0.062	0.023	−0.017	Down	x	NCT
32	0.026	−0.082	−0.084	Down	x	NCT
36	0.035	−0.169	−0.165	Down	x	NCT
41	0.032	−0.021	−0.017	Down	x	NCT
43	0.068	0.016	−0.022	Down	x	NCT
45	0.094	0.045	−0.012	Down	x	NCT
46	0.064	0.004	0.004	Down	x	x
56	0.066	−0.051	−0.021	Down	x	NCT
57	0.055	−0.084	−0.025	Down	x	NCT
302	0.0002	0.496	0.494	Down	x	x
303	0.001	0.394	0.452	Down	x	x
659	0.006	0.038	0.041	Down	x	x
283	−0.032	0.707	0.614	Up	x	x

7. CONCLUSIONS

An evaluation of recent algorithms for voltage sag source relative location has been performed. Their features and the rules used with them have been qualitatively and statistically analysed, and this analysis carried out with real data in order to assess the on-line applicability of the methods. The main contributions of the work are as follows:

- Six fault relative location algorithms have been extensively tested with real data (471 asymmetrical faults) collected by power quality monitors installed on the secondary sides of distribution network transformers.
- The relative fault location of voltage sags caused by single-phase faults have been correctly estimated using PCSC and RCC algorithms, which obtained the highest classification rate for this type of fault. Likewise, the PCSC and DR algorithms can correctly estimate the direction of voltage sag caused by phase-to-phase faults. In any fault relative location scheme, the appropriate algorithm would have to be applied after estimating the type of fault.
- A data mining approach based on MANOVA and CN2 has been used to quantify the significance of features and extract patterns from a huge amount of data. Validation and comparative results have been presented in the ROC space as an extension of confusion matrices, and reveal that similar results can be obtained by observing electrical laws and data mining principles. The same approach could be followed in other power quality projects involving real data.
- A more robust estimation of the sag source has been obtained by combining the PCSC and RS algorithms.

8. NOMENCLATURE

SST	slope of system trajectory
RCC	real current component
DR	distance relay algorithm
RS	resistance sign algorithm
sRS	simplified resistance sign algorithm
PCSC	phase change in sequence current algorithm
MANOVA	multivariate analysis of variance
CN2	CN2 induction algorithm
PQM	power quality monitor

ACKNOWLEDGEMENTS

This work has been partially funded by the Ministry of Education and Science (MEC) under project number DPI2006-09370, grant number FPI (BES-2007-14942), and the energy distribution company of Barcelona, Spain—ENDESA.

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